The Signal in the Geopolitical Noise

Short-Term Conflict Forecasting with Deep Learning and GDELT

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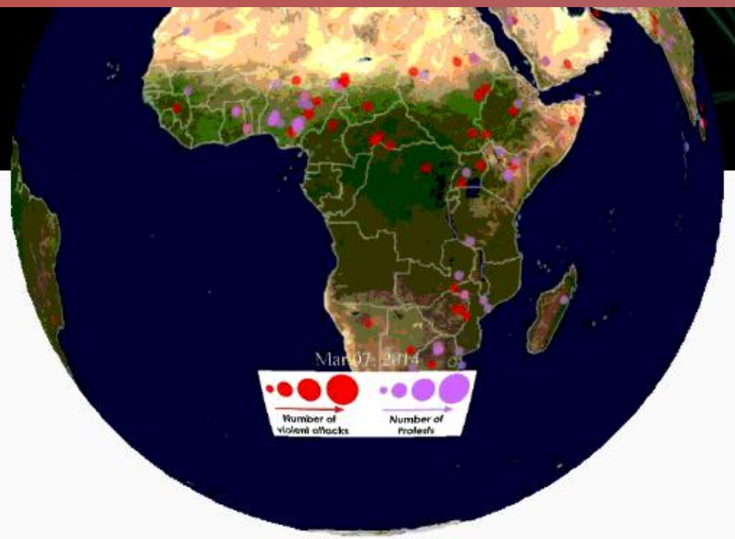
Final Project Presentation

Trained on March 2026 GDELT Data



A Global Database of Society

Supported by Google Jigsaw, the GDELT Project monitors the world's broadcast, print, and web news from nearly every corner of every country in over 100 languages and identifies the people, locations, organizations, themes, sources, emotions, counts, quotes, images and events driving our global society every second of every day, creating a free open platform for computing on the entire world.

The GDELT Project is a realtime network diagram and database of global human society for open research

The Geopolitical Forecasting Problem

The Reality

Geopolitical conflict is highly time-dependent, inherently erratic, and notoriously difficult to predict.

The Resource

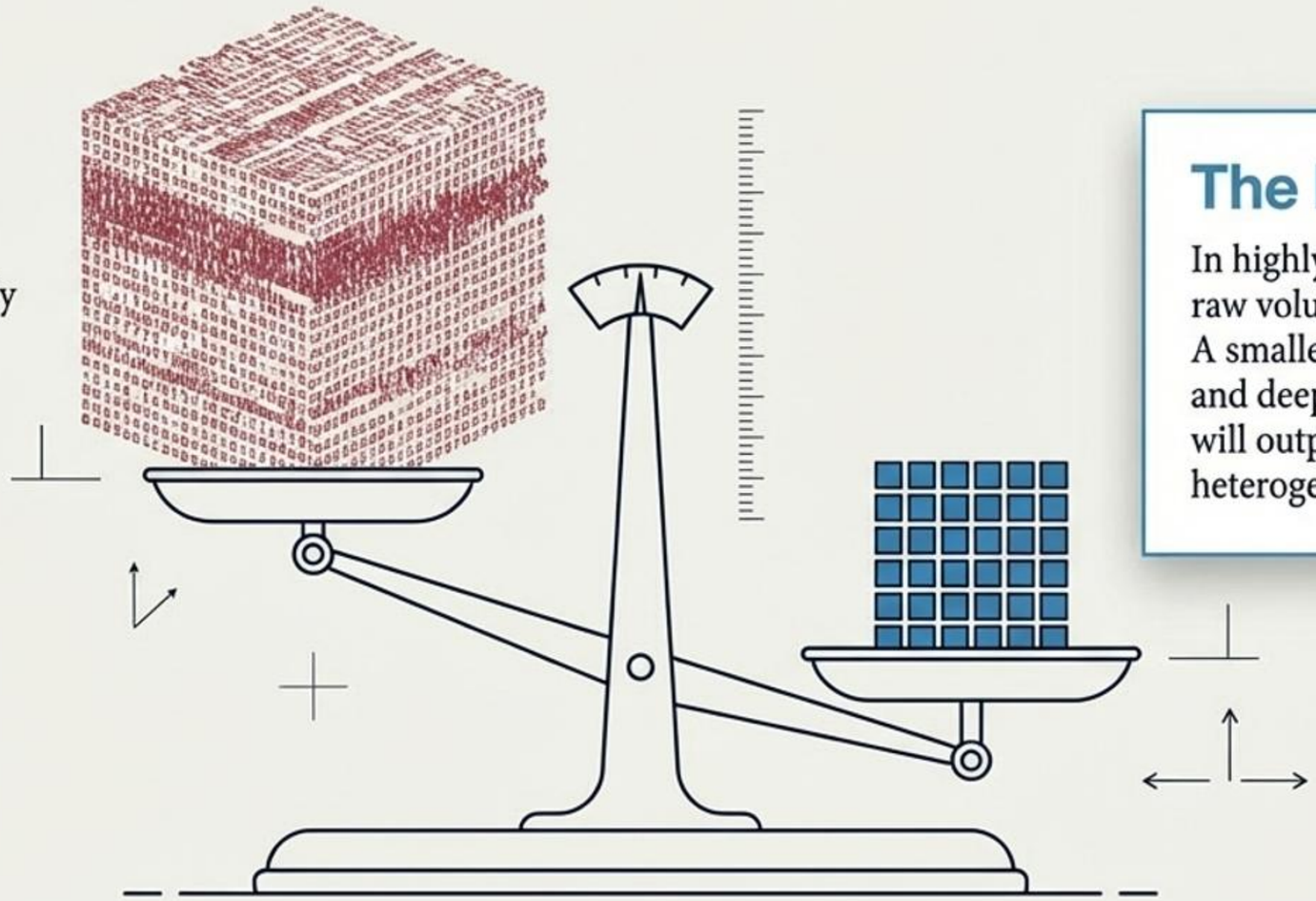
GDELT captures open real-world data from global media and event streams in real-time, creating a massive informational firehose.

Can we use recent GDELT data patterns to accurately forecast short-term Iran-related conflict intensity?

Challenging the Bigger is Better Assumption

The Myth

Feeding a dramatically larger dataset into a deep learning model automatically yields better predictive predictive performance.



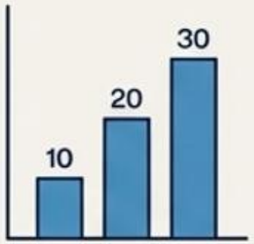
The Hypothesis

In highly specific forecasting, raw volume introduces noise. A smaller, rigidly structured, and deeply relevant subset will outperform a massive, heterogeneous dataset.

Two Distinct Approaches to Data Representation

Raw Downloaded Corpus: 11.399 GiB

The Focused Panel



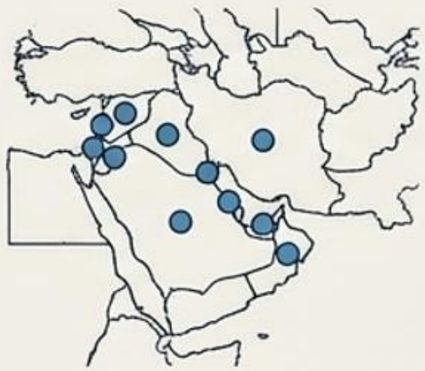
- Contextual regional panel.
- Restricted to 10 key countries (IR, IS, IZ, JO, LE, SA, SY, TU, US, YM).
- Data cleanly aggregated into 15-minute intervals.

The Large-Input Dataset

- Document-level GKG input.
- Contains 81 distinct structural features.
- Expanded to an immense 27.32 GiB (shattering the 5.6 GiB half-size threshold).

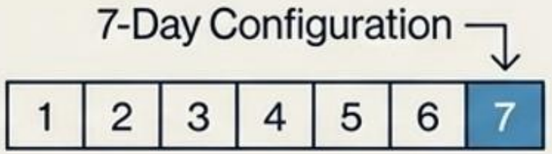
Architecture 1: The Focused Network

Step 1: Contextualize



Filter the global corpus down to 10 deeply relevant regional actors. Preserves vital geopolitical context while stripping away global noise.

Step 2: Sequence



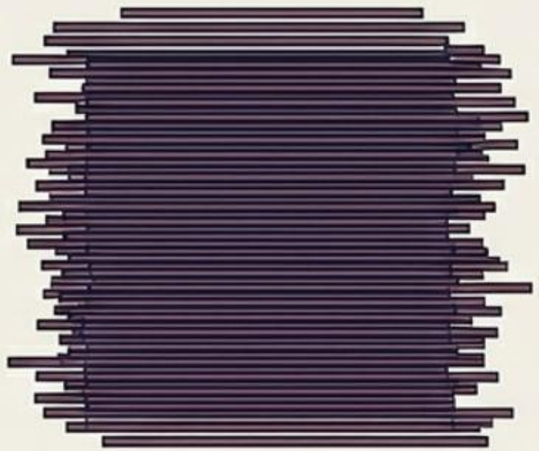
Test 24-hour versus 7-day input windows to capture temporal build-up (7-day configuration proved optimal).

Step 3: Forecast

Deploy a GRU (Gated Recurrent Unit) regression model tailored for time-series data, predicting a strict 1-hour horizon.

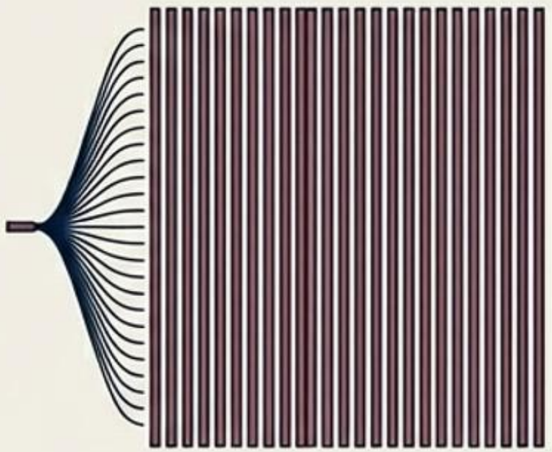
Architecture 2: The Volume Network

Step 1: Maximize Input



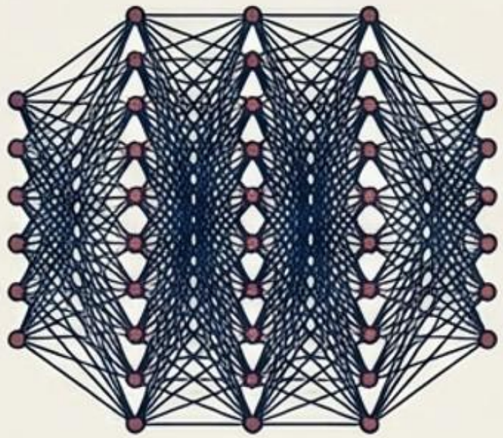
Utilize raw document-level GKG rows, pushing the dataset entirely past the threshold to an immense 27.32 GiB.

Step 2: Expand Features



Ingest 81 distinct structural input features to capture the maximum theoretical breadth across the dataset.

Step 3: Process



Deploy an MLP (Multilayer Perceptron) regressor. Train the system for 5 epochs with a massive 2048 batch size as a strict brute-force comparison.

Predictive Performance Showdown

Focused Network (GRU)

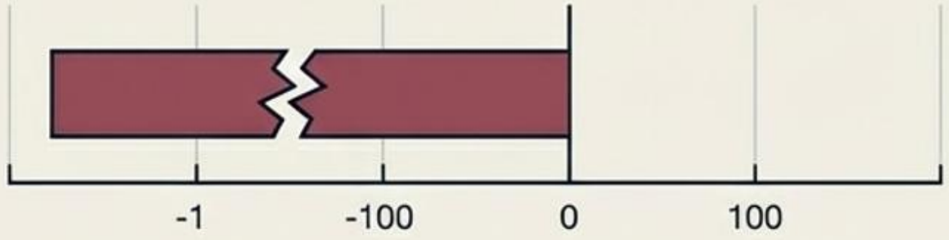
Overall Metrics:
R^2: 0.9368
MAE: 12.4908 | RMSE: 25.1268

Target: Iran-Only Performance
R^2: 0.5190 (Moderate & Meaningful)
MAE: 57.6572 | RMSE: 71.3688

Volume Network (MLP)

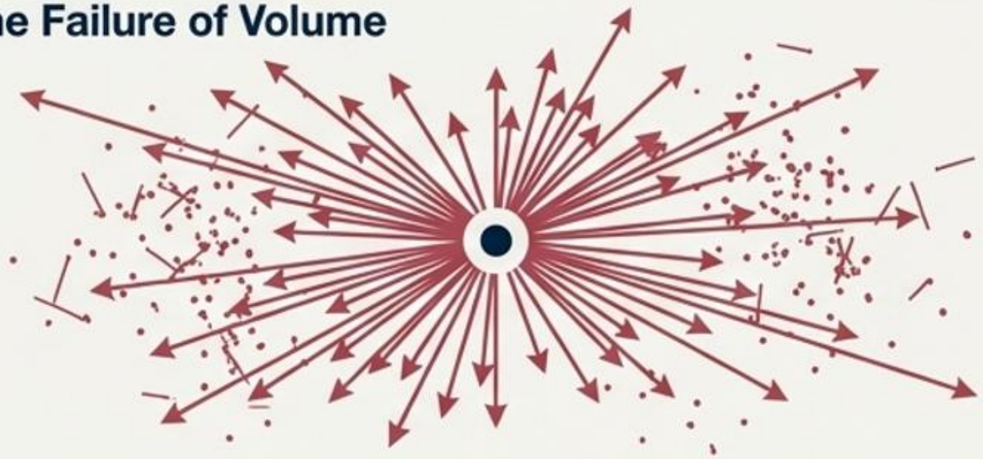
Overall Metrics:
R^2: -1.1476
MAE: 67.7397 | RMSE: 100.4057

Target: Iran-Only Performance
R^2: -1.66e14 (Catastrophic Failure)
MAE: 196.4206 | RMSE: 196.6103



The Anatomy of the Results

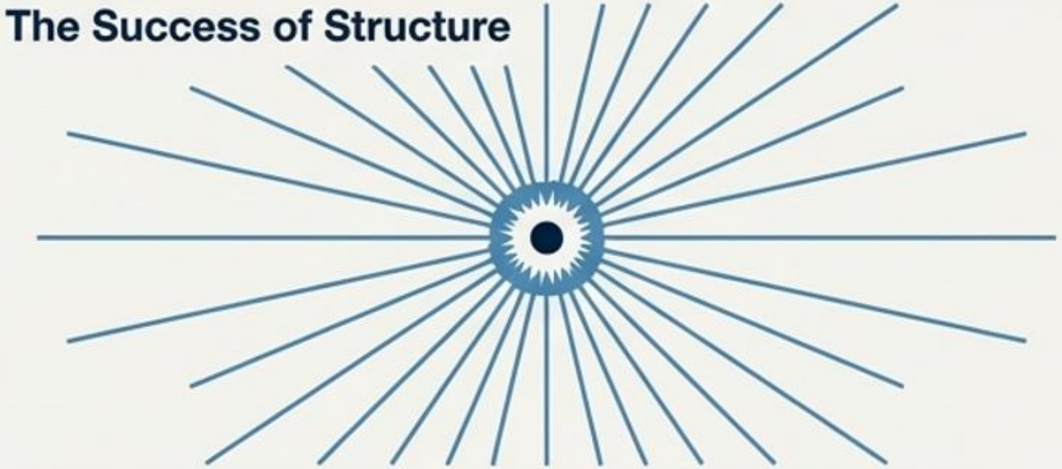
The Failure of Volume



Why did 27.3 GiB fail?

Broad document-level input introduced debilitating noise, heterogeneity, and severe task mismatch. The massive volume actively drowned out the predictive signal.

The Success of Structure



Why did the Focused Panel succeed?

Constraining the problem to a 10-country regional subset preserved critical geopolitical context, allowing the model to actually learn relevant temporal patterns.

The Verdict: The specific representation of the data mattered far more than the sheer volume of the input.

Defined System Boundaries

Data Source

GDELT is a media-coded event system. It strictly tracks the reporting of events across global media, rather than acting as verified battlefield ground truth.

Target Variable

Predictions were based on an engineered future conflict intensity score, rather than a naturally naturally occurring or universally observed label.

Temporal Scope

The findings and trained weights are strictly bounded to the complex geopolitical dynamics isolated specifically within the March 2026 data period.

The Signal Found

1 The Optimal Setup:

A focused, 10-country regional panel feeding a GRU with a 7-day lookback provides meaningful ($R^2 = 0.51$) short-term conflict forecasting.

2 The Myth Busted:

Forcing a strictly larger dataset (27.3 GiB) into the system actively degraded predictive performance, proving that in this context, bigger is worse.

3 The Ultimate Lesson:

In deep learning for geopolitics, precise subset definition, task alignment, and structural relevance are exponentially more valuable than raw data volume.